MIL-S-37695
16 August 1978

MILITARY SPECIFICATION

SULISOBENZONE LOTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Sulisobenzone Lotion (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6505

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE.
FOOD AND DRUG ADMINISTRATION (FDA).

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204).

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Applications for copies should be addressed to the Mack Publishing Company, Easton, PA 18042).

3. REQUIREMENTS

3.1 Material. Shall be a lotion containing 10.0 percent sulisobenzone and 0.7 percent preservatives (as parabens), by weight, in a suitable lotion base.

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3.1.1 The finished lotion may be prepared from an intermediate aqueous solution of sulisobenzone, providing the sulisobenzone powder used in the preparation is tested and meets the requirements of the specification for sulisobenzone (see 3.2.1). The water used in the preparation of the solution shall be tested and shall meet the requirements of the USP for Purified Water.

3.1.2 Assay limits.

3.1.2.1 Sulisobenzone. The finished preparation shall assay to contain not less than 90.0 percent and not more than 110.0 percent of the required amount of sulisobenzone, when tested as specified in 4.7.1.

3.1.2.2 Total parabens. The finished preparation shall assay to contain not less than 90.0 percent and not more than 110.0 percent of the required amount, when tested as specified in 4.7.2.

3.1.3 Fungi, yeasts, or molds. Shall contain no fungi, yeasts, or molds. The bidder or offeror shall submit with its bid or proposal the method (s) used to determine the absence of fungi, yeasts, or molds, which shall be subject to approval and acceptance by the Government. If the method(s) are approved and accepted by the Government, these method(s) shall be employed in the testing of the material. The requirement for submission of the method(s) may be waived, provided the offeror states in his bid or proposal, that the product he proposes to furnish is the same product he has offered to the purchasing activity on a previous procurement, and the contracting officer determines that such product was previously procured and/or tested by the purchasing activity and found to comply with the requirement for absence of fungi, yeasts, or molds.

3.2 Ingredients.

3.2.1 Sulisobenzone. The sulisobenzone used in the manufacture of the final preparation shall be of the highest pharmaceutical grade and shall comply with the following:

3.2.1.1 Assay. The sulisobenzone powder shall assay not less than 99.0 percent when tested as specified in 4.7.3.

3.2.1.2 Identification. The infrared absorption spectrum (graph) of sulisobenzone shall compare qualitatively with the spectrum of sulisobenzone standard as kept on file by the manufacturer. A copy of the standard infrared curve shall also be made available to Government upon request.

3.2.1.3 Melting point. Melting point of the powder shall be not less than 145°C with decomposition, using USP method, Class I.

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3.2.2 Propylparaben. The propylparaben used in the manufacture of the finished preparation shall be in accordance with tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.2.3 Methylparaben. The methylparaben used in the manufacture of the finished preparation shall be in accordance with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.3 Other ingredients. All other ingredients used in the manufacture of the finished product shall be of USP or NF quality, or if not included in either of these compendia, the ingredients shall be of the highest pharmaceutical grade and suitable for use in a lotion to be employed on the human skin.

3.4 Expiration dating period. The expiration dating period shall be not less than 36 months.

3.4.1 Delivery. Not more than 3 months of the expiration dating period shall have elapsed at the time of delivery to the Government.

3.5 Workmanship. The material and its containers shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

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4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of lotion produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

TABLE I. Sampling.

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Filled immediate container
Minor	II	2.5	Filled immediate container

4.3.1 Raw material and end item testing. Noncompliance with any of the following is cause for rejection.

TABLE II. Criteria for raw material and end item testing.

Characteristic	Requirement paragraph	Inspection level 1/	Minimum sample
Assay for Sulisobenzone	3.1.2.1	S-2	5 } composite
Assay for Total Parabens	3.1.2.2	S-2	5 } composite
Ingredients 2/ Sulisobenzone	3.2.1		

1/ Where possible, the sample shall be used for two or more tests.
2/ Sufficient quantities of each of the raw materials shall be selected from each lot to perform tests indicated in applicable test methods. Failure of any sample to comply with test requirements shall be cause for rejection of the lot represented.

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4.4 Examination. The finished lotion shall be examined to determine compliance with all requirements of this specification. Non-conformance will be permitted to the extent indicated in 4.3 and 4.3.1.

4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

TABLE III. Classification of defects.

Categories	Defects 1/
Major	
101	Lotion not free from foreign matter.
102	Lotion has crystal formation.
103	Lotion not homogeneous.
Minor	
201	Exterior of immediate container is not free from contents or foreign matter.
202	Label not free from defacing marks or tears.

1/ Inspection is not restricted to the classified possible defects listed above.

4.5 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.6 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.7 Assays.

4.7.1 Assay for sulisobenzone in final product.

Procedure.

Accurately weigh approximately 500 mg of sample into a 50 ml volumetric flask. Dissolve in reagent grade absolute methanol and dilute to volume. Shake thoroughly and filter through

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Whatman No. 1 filter paper. Pipet 5.0 ml of clear filtrate into a second 50 ml volumetric flask, dilute to volume with methanol and mix. Pipet 5.0 ml of this dilution into a third 50 ml volumetric flask, dilute to volume with methanol and mix. Accurately weigh approximately 50 mg of sulisobenzone reference standard and dilute stepwise with methanol in the same manner as for the sample.

Determine the absorbances of the sample and standard preparations at 285 nm, in a 1 cm cell, using a suitable spectrophotometer and anhydrous methanol as the reference blank.

Calculation.

$$\text{Percent sulisobenzone} = \frac{A_u \times W_s \times 100}{A_s \times W_u}$$

Where:

A_u = the absorbance of the sample preparation
A_s = the absorbance of the standard preparation
W_s = weight of the standard in milligrams
W_u = weight of the sample in milligrams

4.7.2 Assay for total parabens.

Standard preparation.

Accurately weigh 500.0 mg of methylparaben and 200.0 mg of propylparaben and dissolve in anhydrous methanol in a 200 ml volumetric flask. Dilute to volume with anhydrous methanol and mix. Pipet 2 ml of this solution into a 100 ml volumetric flask, dilute to volume with anhydrous methanol and mix. Pipet 10 ml of the above solution into a 100 ml volumetric flask, dilute to volume with anhydrous methanol and mix.

Procedure.

Accurately weigh approximately 1 gram of the final product into a suitable container and transfer quantitatively to a 125 ml separatory funnel with not more than 10 ml of purified water. Extract with five 25 ml portions of chloroform, filtering each extract through a small piece of cotton into a 250 ml beaker. Evaporate the chloroform on the steam bath to about 10 ml, remove from the steam bath and pass nitrogen into the beaker until the

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odor of chloroform is no longer perceptible. Add 50 ml of anhydrous methanol to the beaker and heat until the residue is evenly dispersed in the methanol. Transfer the methanol quantitatively to a 100 ml volumetric flask, rinsing the beaker with several small portions of methanol, and transferring the rinsings to the flask. Cool to room temperature, dilute to volume with methanol and mix. Filter through Whatman No. 1 paper. Reject the first 20 ml and pipet 10 ml of the remaining filtrate into a second 100 ml volumetric flask, dilute to volume with methanol and mix.

Determine the absorbance of the sample and standard solutions at 255 nm using a suitable spectrophotometer and anhydrous methanol as the reference blank.

Calculation.

$$\text{Total parabens (as 5:2; methyl; propyl)} = \frac{Au \times 0.0007 \times 100 \times 100}{As \times W_u \times 10}$$

where:

Au = the absorbance of the sample preparation
As = the absorbance of the standard preparation
Wu = weight of sample in grams

4.7.3 Assay of sulisobenzone active ingredient.

Procedure.

Accurately weigh approximately 50 mg of active ingredient powder into a 50.0 ml volumetric flask and dissolve in reagent grade absolute methanol and make to volume. Shake thoroughly and filter through Whatman No. 1 paper. Pipet 1.0 ml of the filtrate into a 100 ml volumetric flask and made to volume with methanol and mix. Accurately weigh approximately 50 mg of sulisobenzone reference standard and dilute stepwise with methanol as done for the sample.

Determine the absorbance of the sample and standard preparations at 285 nm, using a suitable spectrophotometer and anhydrous methanol as the reference blank.

Calculation.

$$\text{Percent sulisobenzone} = \frac{A_u \times W_s \times 100}{A_s \times W_u}$$

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where:

A_u = the absorbance of the sample preparation
A_s = the absorbance of the standard preparation
W_u = the weight of the sample in milligrams
W_s = the weight of the standard in milligrams

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One bottle containing 75 grams, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available immediate container considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. At the option of the contractor, a unit package may be supplied.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. Twelve (12) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Twelve (12) units of issue shall be packed in a substantial exterior shipping container of the type, size and kind commonly used for this purpose.

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5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder, and shall include the information required below:

5.4.1.1 Immediate container: Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the expiration date.

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the National Stock Number.

NOTES: (1) The National Stock Number need not be preceded by any prefix.

(2) Strip labeling may be used.

(3) Label adhesion test shall not apply.

(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

5.4.2 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall also be shown.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

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(a) Title, number and date of this specification.

(b) National Stock Number.

(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item:

National Stock Number	Item Identification
6505-00-443-4616	SULISOBENZONE LOTION, Ultraviolet Screen, 10%, 75 Grams

Custodians:

Army - MD
Navy - MS
Air Force - 03

Review activity
DoD - MB

Preparing activity:
DLA-DM
Project No. 6505-1840

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

SULISOBENZONE LOTION MIL-S-37695

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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